BSc Thesis

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TROPOSPHERIC OZONE AND ITS EFFECT ON THE GROSS PRIMARY PRODUCTIVITY OF THE LOOBOS PINE FOREST

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Abstract

Tropospheric ozone (O_3) is an atmospheric pollutant known to impair plant productivity by disrupting photosynthesis. This study investigates the effect of ozone exposure on the gross primary productivity (GPP) of the Loobos pine forest in the Netherlands. Using two years (2023–2024) of observational data, including ozone concentrations and key environmental variables such as temperature, vapor pressure deficit, and radiation, light response curves (LRCs) were generated across a range of ozone levels. To isolate ozone's impact, the analysis was restricted to conditions near the photosynthetic optimum. GPP was assessed via the maximum photosynthetic rate (Amax), revealing a clear negative relationship with ozone: Amax declined by nearly 50% as ozone concentrations increased from 20–30 μg/m^3 to 110–120 μg/m^3. These results reflect acute, short-term responses primarily driven by ozone-induced reductions in stomatal conductance. Such reductions in photosynthetic capacity, even over short timescales, weaken the terrestrial carbon sink.

Introduction

1.1 Background

Since the start of the industrial revolution, the emission of greenhouse gases (GHGs) has risen dramatically, primarily due to an increase in the burning of fossil fuels. This has become the leading driver of human-induced global warming, posing one of the greatest challenges humanity faces today. Scientists are observing climate shifts that are unprecedented over hundreds of thousands of years. These changes could trigger tipping points, leading to irreversible disruptions in the equilibria in Earth's natural systems (Steffen et al., 2015). The concentration of GHGs in the atmosphere depends not only on the emissions but also on natural absorption. The buffer capacity of land and oceans plays a crucial role in redistributing the anthropogenic CO_2 emissions from the atmosphere to the ocean and terrestrial biosphere. These processes are part of the carbon cycle. According to the latest data, 37.4 $GtCO_2$ was emitted in 2024. Approximately 55% of those emissions are taken up by oceans and the terrestrial biosphere, with oceans taking up 25% and the terrestrial biosphere absorbing 30% (Friedlingstein et al., 2025). In terrestrial ecosystems, vegetation's ability to absorb carbon is determined by gross primary productivity (GPP)—the amount of carbon taken up through photosynthesis. GPP depends on various environmental factors including temperature, radiation, soil water content, relative humidity, nitrogen availability, and CO_2 concentration, which can enhance or constrain photosynthetic activity, depending on their levels (Fleischer et al., 2013, Xu & Chen, 2024). There are also atmospheric pollutants which affect GPP. One of these pollutants is ozone (O_3). High ozone concentrations can damage plants, disrupting photosynthesis (and thus GPP) but also impair their ecosystem services.

Ozone concentrations are expected to remain above pre-industrial levels due to NOx (Nitrogen Oxides) and VOC (Volatile Organic Compound) emissions, meaning its impact on vegetation will persist in regions with high ozone concentrations. This, in turn, reduces the global carbon sink, affecting the carbon balance. To what extent this will happen remains uncertain, especially since ozone's effect on GPP is vegetation specific. In my thesis, I aim to assess how ozone will affect Loobos, a pine forest in the Netherlands.

1.2 Research questions

To substantiate the research and provide a structured approach, the following research questions have been formulated:

Main research question: To what extent does tropospheric ozone exposure affect the gross primary productivity of the pine forest at Loobos?

Sub-research questions:

- What are the spatial and temporal variations of ozone levels in the region?
- What are the chemical mechanisms through which ozone exposure affects vegetation?
- Is there a threshold ozone concentration above which GPP shows a significant decline?

1.3 Ozone pollution

This chapter explores the role of ozone as a pollutant, which depends on which layer of the atmosphere it is present in. The lower atmosphere consists of the troposphere, the tropopause and the stratosphere. The troposphere is the lowest part of the Earth's atmosphere, starting at the surface and ending at the tropopause, which varies in height depending on latitude and time of year, but generally lies between 10 and 15 kilometers above sea level. The stratosphere extends from the tropopause to the stratopause, which is between 45–55 kilometers (Seinfeld & Pandis, 2012). Around 90% of all ozone can be found in the stratosphere (Lelieveld & Dentener, 2000), where it plays a crucial role in absorbing harmful UV-radiation. The remaining 10% of ozone is found in the troposphere, where background concentrations are increasing. Janach (1989) states that an analysis of ozone measurements made in mostly rural European sites indicates a 1–2% annual increase in average concentrations over the past 40 years. In the troposphere, an increase in ozone concentration is not positive, as it acts as a greenhouse gas, is harmful to human and animal health, and negatively affects vegetation and agricultural yields.

1.3.1 What causes tropospheric ozone pollution

Tropospheric ozone depends on two processes: stratosphere–troposphere exchange (STE) and photochemical production from precursors, with the latter being dominant (Seinfeld & Pandis, 2012).

The photochemical formation starts with NO_2 photolysis:

$$\mathrm{NO_2 + h\nu \rightarrow NO + O(^3P)} \tag{1.1}$$
$$\mathrm{O(^3P) + O_2 \rightarrow O_3} \tag{1.2}$$

Once this reaction commences, it causes another chain reaction which can be divided into 3 categories: initiation, propagation, and termination. These are the simplified reactions that take place in each step:

Initiation:

- O_3 + hν $\rightarrow$ O(^{1}D) $\rightarrow$ OH formation
- HCHO photolysis $\rightarrow$ radicals
- Alkene + O_3 $\rightarrow$ radicals

Propagation:

- OH + VOC $\rightarrow$ RO_2
- RO_2 + NO $\rightarrow$ NO_2 + OH

Termination:

- OH + NO_2 $\rightarrow$ HNO_3

(Kleinman, 2004, Lelieveld & Dentener, 2000)

This underscores the roles of NOx and VOCs as precursors for tropospheric ozone formation. Therefore, if we want to be able to predict or analyse ozone concentrations, it is crucial to also look at these emissions. After the photolysis of NO_2, an equilibrium is typically established between the reaction which forms ozone, and the reaction by which ozone reacts with NO to form NO_2 and O_2.

However, the presence of VOCs alters this balance. VOCs react with hydroxyl (OH) radicals to form peroxy radicals (RO_2•), which in turn react with NO. This reaction removes NO without consuming ozone, thereby disrupting the equilibrium and allowing ozone to accumulate in the atmosphere.

This net accumulation of ozone depends on the relative availability of NO_x and VOCs. If either of these precursors are significantly less abundant than the other, it becomes the limiting factor in ozone formation (Nguyen et al., 2022). The majority of NOx originates from anthropogenic sources. There are also natural emissions, including lightning and denitrification processes, although human activities, particularly combustion processes, are the dominant cause. The main anthropogenic sources of nitrogen oxides are road transport and the public electricity and heat sector, contributing 39.4% and 18.7% respectively (European Environment Agency, 2024).

The emissions from developed countries have been steadily declining in recent decades as a result of newer technologies and better management policies. However, this reduction in NO_x does not necessarily lead to a corresponding decrease in ozone concentrations. In urban areas where ozone formation is VOC-limited, reducing NO_x can actually lead to higher ozone levels because less NO is available to titrate ozone (Zhao et al., 2019). It is also important to note that ozone concentrations are typically much higher than the concentrations of its precursors, and the relationship between precursor emissions and ozone levels is non-linear. Small shifts in the NO_x-to-VOC ratio can result in large changes in ozone production (Zhao et al., 2022).

1.3.2 Temporal and spatial variability of ozone

To be able to explain the reason for high or low ozone concentrations, the cause of ozone formation alone is not enough. Tropospheric ozone is both temporally and spatially heterogeneous, which complicates accurate projections of current regional trends or future concentrations (Wittig et al., 2008). Since ozone production is driven by sunlight, it follows a distinct diurnal cycle. Therefore, ozone concentrations are not only location-dependent but also vary throughout the day. It also means that net ozone accumulation happens only during daylight hours. During the night, there is a net loss of ozone, if NO is present. At sunrise, ozone concentrations are typically low because no ozone is formed during the night. As traffic increases around this time, the concentration of NO_2 in the air increases as well. Near the source, this causes a decrease in ozone, because the traffic NOx emissions consist of NO for about 80–90% (Grange et al., 2017), and this shifts the equilibrium reaction described in section 1.3.1 towards consuming ozone. Since ozone is only consumed near the source, and since the reaction that consumes ozone produces NO_2, there will eventually be a net accumulation of ozone because of these emissions of NOx. Another factor contributing to the diurnal variation is the boundary layer. The boundary layer is the lowest part of the troposphere, and its height varies depending on surface heating and atmospheric stability. During the day, the boundary layer grows and vertical mixing increases. This brings ozone-rich air from higher in the troposphere to the surface (Stull, 2018). The peak ozone concentration is usually around the mid to late afternoon, when sunlight is most intense. In the evening, the photochemical reaction slows down as the amount of sunlight decreases. Eventually, ozone production halts completely, and the ozone that remains gets gradually depleted due to deposition or by reacting with NO in the atmosphere. As a result, ozone concentrations drop significantly, often reaching very low levels, or at times even approaching zero.

Although this diurnal pattern generally persists, many processes can vary substantially in strength and location throughout the day. Therefore, there is a considerable variation in day-to-day surface ozone levels at any particular location (Liao et al., 2022). Other than day-to-day variation, ozone levels also differ greatly depending on the season. The reason for this is quite simple, considering the production of ozone is a photochemical reaction. During summer there is more sunlight, causing both a higher rate of ozone production because of the stronger radiation, and a longer timeframe of ozone production, since the sun rises earlier and sets later compared to other seasons.

So far, it has been described that many factors contribute to the concentration of ozone. But beside sunlight and the emissions of precursors, there are still some other meteorological factors to take into account. An illustration of these processes can be seen in annex A. Firstly, temperature has a positive correlation with ozone concentration. This is because higher temperatures tend to be caused by the abundance of sunlight, and higher temperatures also enhance the rate of the photochemical reactions leading to the formation of ozone. It also promotes the emission of VOCs (Coates et al., 2016). Humidity, in turn, negatively affects ozone due to increased wet deposition of both ozone and ozone precursors. The same is true for rainfall. Mixing of the atmosphere, both horizontal due to wind and vertical due to pressure differences, disperses ozone. When looking at a specific location this could mean either a positive or negative effect. High pressure systems trap pollutants near the surface, leading to a higher concentration there. Low pressure systems have the opposite effect, and similarly, wind can either transport ozone-rich air to a location or bring in cleaner air, depending on the source region. These are just the direct effects on ozone, but these meteorological factors can also have a compounding effect and can also affect each other. For example, a high-pressure system occurs when temperatures are higher at the surface, meaning that under these circumstances both the rate of ozone formation increases, and these concentrations are then also trapped. On the other hand, temperature also has a positive effect on humidity, which in turn has a negative effect on ozone concentrations. To summarize, the concentration of ozone in the lower atmosphere is the result of a complex interplay between emissions, chemical processes, and meteorological conditions. While the availability of sunlight and precursors like VOCs and NO_x is essential for ozone formation, factors such as temperature, humidity, atmospheric pressure systems, wind patterns, and precipitation significantly influence the net outcome. And yet, there are still other factors at play that have not been explained, such as the stratosphere-troposphere exchange mentioned at the start of the chapter. Although this also plays a role, it was decided that these processes will not be explained in detail to avoid going off topic.

Despite the complexity of the temporal and spatial variability of ozone, there are some patterns and trends that hold most of the time. An important difference is the background concentration of ozone in rural and urban areas. Rural concentrations in Europe are on average around 1 ppb higher (Nguyen et al., 2022b). The highest ozone concentrations are typically found in urban plumes around 50–100 km from the city center (Sillman, 1999). Rural areas also often have higher emissions of VOCs. The location of Loobos is relatively rural, and not too far from major urban centers, meaning the background concentration is often relatively high.

1.4 Deposition of ozone to vegetation

1.4.1 How does ozone enter the stomata

Ozone is removed from the troposphere via chemical reactions or atmospheric deposition. Vegetation plays an important role in the deposition of ozone. Ozone can be taken up by vegetation via the stomatal or non-stomatal pathway. The stomatal pathway means ozone enters the stomates. In contrast, non-stomatal uptake occurs through various other routes, including deposition to the external leaf tissue, bark, solution in water or organic matter on top of the soil (ICP Vegetation et al., 2012).

The contribution of non-stomatal ozone loss depends on vegetation type, season, and weather conditions. Leaf area also strongly influences the balance between stomatal and non-stomatal removal. Furthermore, removal rates vary within the plant canopy, driven by factors such as air turbulence and the physiological status of the leaves.

Although the ratio between non-stomatal and stomatal uptake may differ, modelling studies have shown that the removal of ozone through non-stomatal pathways is not sufficient to reduce the flux of ozone through stomata and so cannot serve as a protective mechanism for reducing leaf uptake of ozone. This is because outer surfaces have lower and more limited uptake rates, making the stomatal aperture the dominant control when surface resistances differ. As a result, even small stomatal openings can sustain significant ozone flux under typical diurnal conditions (Altimir et al., 2008). Ozone can also be removed by chemical reactions with NO emissions from soils and VOCs emitted by vegetation, but these are typically considered atmospheric chemical sinks, not part of non-stomatal uptake. At the leaf level, however, VOCs emitted by some plant species can react with ozone near the leaf surface, potentially reducing the amount of ozone reaching stomata and thus offering some protection against ozone-induced damage.

1.4.2 How does damage occur in the plant

When ozone enters the leaf, it forms reactive oxygen species (ROS) by interacting with the cell membranes and walls of the sub-stomatal pores. This initiates a chain reaction of ROS production in adjoining cells. Plants possess antioxidants such as ascorbic acid that help scavenge ROS, and detection of ROS in guard cells also triggers a signalling cascade that leads to stomatal closure via ion channel regulation, serving as a short-term protective mechanism to limit further ozone entry (ICP Vegetation et al., 2012, Vahisalu et al., 2010). However, the plant's detoxification capacity is limited, and when ozone exposure exceeds this capacity, the defence mechanisms become overwhelmed. As a result, free-radical–induced cell death occurs, further stomatal closure. The photosynthetic machinery becomes less efficient due to reduced Rubisco and chlorophyll content. Visible symptoms such as interveinal necrosis, stippling, and early leaf senescence often follow, with damaged leaves falling from the plant prematurely. This accelerated loss of leaf area reduces the plant's photosynthetically active surface, compounding the reduction in GPP.

Ozone's impacts on GPP manifest on both immediate and longer-term timescales. On the one hand, there are direct, short-term effects that reduce GPP almost instantly upon ozone exposure, such as reduced stomatal conductance, and damage to photosynthetic proteins. On the other hand, ozone also leads to compounding long-term effects, including early leaf senescence, reduced leaf area index, altered carbon allocation, and changes in root-to-shoot ratios, which all further diminish GPP over time (Blokhina, 2002).

1.4.3 Which factors determine how a plant is affected

The rate of ozone removal through vegetation depends on numerous factors, which are related to both atmospheric and vegetative characteristics. Firstly, different plant species possess different stomatal conductivities. Species with naturally higher stomatal conductance allow more ozone to enter the leaf, leading to greater internal exposure and damage. This has been linked to higher ozone sensitivity in species like wheat, rice, and white clover. Conversely, species with lower stomatal conductance tend to be less affected. The ability of a species to neutralize reactive oxygen species (ROS) generated by ozone exposure also determines sensitivity. Some species have inherently stronger detoxification mechanisms.

There is variability not just between species, but also between different varieties or provenances (local genotypes) of the same species. These genetic differences can influence both stomatal behaviour and detoxification potential. Pine trees, as part of the needle leaf group, tend to be less sensitive to ozone in terms of GPP reductions. Studies (e.g., Wittig et al., 2009) indicate that needle leaf trees like spruce (and likely pine) show minimal declines unlike broad leaf species such as beech, which can suffer major reductions in growth and carbon sequestration. This suggests pines may have a competitive advantage in mixed forests as ozone levels rise.

1.5 Research set-up

The preceding sections have shown that the impact of ozone on vegetation is determined by a complex interplay of atmospheric chemistry, meteorological variability, plant physiological traits, and species-specific sensitivities. While broad patterns have been established through experimental studies and modelling efforts, the actual effect of tropospheric ozone on GPP in natural forest systems remains difficult to quantify, particularly under real-world environmental conditions.

The objective of my research is to apply available knowledge, along with existing local data of ozone concentrations, and other variables which will be mentioned in the methodology, to estimate the extent to which Loobos is affected by ozone, focussing on the forest's GPP. To achieve this, 2 years (2023 & 2024) of local observational data will be used, and from this, a dose-response relationship will be established.

Methodology

To conduct this research, relevant environmental data were collected and processed. Ozone concentration data were obtained from the Wekerom-Riemterdijk monitoring station, located approximately six kilometers from Loobos (see annex B). This station is part of the National Air Quality Monitoring Network, and it provides measurements of background ozone levels. Hourly average ozone concentrations were retrieved for the years 2023 and 2024 from Luchtmeetnet.nl. In addition, data for GPP and for variables which affect GPP were collected, which are air temperature, vapor pressure deficit (VPD), global radiation (Rg), and soil water content (SWC). Loobos is a research site located in a pine forest in the Veluwe natural park, representing a large area dominated by pine trees. The forest was planted in 1910, and the research site was established in 1995. Data for these variables were sourced from the Integrated Carbon Observation System (ICOS) data portal, also for 2023 and 2024 (link). ICOS provides half-hourly data, but these values were then averaged to hourly averages to align temporally with the ozone concentration data.

Once the data were obtained, all variables required for constructing light response curves were available. An initial light response curve was generated using all hourly GPP averages from 2023 and 2024 (figure 1). To more effectively isolate the effects of ozone on photosynthetic activity, a narrow range of optimal environmental conditions for GPP was defined. While literature reports a broad range of temperature optima for GPP, with the ecosystem-scale optimum air temperature for GPP, T$_{o}pt^{e}co, averagingaround236Cdependingonvegetationtypeandregion(Huangetal., 2019), thetempera$
$-22\,C.Thisnarrowerwindowwasintentionallychosentominimizetheinfluenceoftemperaturevariabilitya$

The optimal VPD was difficult to identify, as the relationship between GPP and VPD remains uncertain. Huang et al. (2024) reports that GPP reaches its maximum at a vapor pressure deficit of roughly 0.8–0.95 kPa, with productivity declining once VPD exceeds this threshold. Lastly, GPP has a hump-shaped response curve to SWC. The optimal range depends on soil characteristics. For sandy soils such as those present at Loobos, the optimum likely occurs in the 8–12% range. These ranges were ultimately not applied since these strict criteria resulted in too few data points to produce a meaningful analysis. Therefore, the VPD range was broadened to 0–15 kPa, and SWC was excluded from the selection criteria because in most non-drought conditions, the trees have access to the groundwater.

The next step involved generating new light response curves (LRCs). GPP was compared across different ozone ranges [0–10, ... 160–170] while keeping the other variables constant at their defined optimal values. The maximum photosynthetic rate (Amax) was then determined for each of the ozone concentration ranges. Concentrations above 120 μg/m^3 were too uncommon within the 20–22 °C range to be used.

For the higher ozone concentration ranges (>50 μg/m^3), the Amax values could be derived directly from the data. This is because higher ozone levels correspond with higher light intensities, and higher light intensities have a positive effect on GPP. However, for lower ozone concentrations, GPP values were generally low due to the associated low levels of sunlight, making direct estimation of Amax challenging. To address this, a model fit was applied. A threshold of 500 W/m^2 was set, meaning that if there were plenty of datapoints where Rg exceeded this value, the Amax was taken directly as described above. If not, it was fitted.

A simplified light response model was chosen in the form:

$$GPP = A_{\max} \cdot (1 - e^{-k \cdot Rg})$$

where k represents the light-use efficiency at low irradiance. The parameters Amax and k were estimated by minimizing the residual sum of squares (RSS) between the observed and modelled GPP:

$$RSS = \sum_{i} [GPP_{observed,i} - GPP_{model,i}(A_{\max}, k)]^2$$

The resulting fits were visualized through plots. If there were fewer than five available datapoints within one ozone range, this range was skipped, and no Amax value was given. It should be noted that these estimates involve considerable uncertainty, which will also be described in the discussion.

Finally, a trendline was plotted across the different Amax values for the ozone ranges, and the slope of this trendline can be used to estimate the effect of ozone on GPP.

Results

In figure 2, the dots show each measurement of GPP made during the selected period of 2023-2024 as a function of light intensity, with the colour being dependent on the ozone concentration. Using these values, light response curves were plotted under different ozone concentration ranges while keeping temperature around the optimum (20-22°C) and VPD between 0-15 kPa. The higher ozone ranges resemble the typical asymptotic shape of a light response curve, having a high initial light use efficiency, and flattening off at higher light intensities. For the lower ozone ranges, there was an insufficient amount of data for higher light intensities. This is why, when looking at the 20-40 range for example, which is the orange line in figure 2, the curve continues until around 300 W/m2, after which it abruptly stops. This means that there were not enough measurements where ozone concentration was between 20-40 µg/m3 and light intensity was above 300 W/m2. The opposite is true for the high ozone range, where there were not enough measurements of ozone being 100-120 µg/m3 and light intensity being lower than 200 W/m2.

The most significant pattern made visible by this figure, however, is the difference in GPP per ozone range. Lower ozone ranges show a steeper initial slope and higher Amax, while higher ozone concentrations show a smaller slope and lower Amax.

From these results, the Amax values were taken to estimate the difference in Amax for the different ozone ranges. Figure 3 is used to assist in clarifying the need to use a fit for lower ozone concentrations. It is a box plot of the frequency of light intensities for each ozone concentration range. As can be seen, as light intensity increases, ozone concentrations on the lower end stop occurring.

The Amax plotted against ozone concentrations can be seen in figure 4. It shows a significant downward trend, with an almost 50% decrease in Amax when ozone concentrations increase from 20-30 µg/m3 to 110-120 µg/m3. The error bars represent the standard deviation for each range.

Discussion

The results indicate a notable negative relationship between ozone concentration and GPP, with figure 4 indicating a reduction of approximately 50% in GPP observed at ozone levels of 110–120 µg/m³ compared to the 20–30 µg/m³ range. This negative correlation between ozone and GPP can also be seen in figure 1 and coincides with existing literature. However, the magnitude of reduction is notable. The study conducted by Wittig et al. (2007) concluded that the elevation of ozone concentrations compared to before the industrial revolution caused an 11% decrease in Amax.

There are several key methodological differences that are likely to explain the discrepancy in results. First, Wittig et al. (2007) looked at an average ambient ozone concentration of 94 µg/m³ compared to charcoal-filtered air in which ozone concentrations are below 20 µg/m³. This range is similar to the range used in this study, but even slight deviations can cause different results. Second, Wittig et al. (2007) averaged responses across multiple species, while this study focused solely on Scots pine. However, this goes against what is expected. As explained in section 1.4.3, pine trees tend to be less sensitive to ozone in terms of GPP reductions. The fact that this study observed a much larger GPP reduction in Scots pine than the multi-species average reported by Wittig et al. (2007) suggests that either the local pine population is unusually sensitive to ozone, or that there is another methodological difference.

Third, Wittig et al. (2007) quantified ozone exposure using cumulative ozone uptake (CU), a physiological measure that accounts for both concentration and stomatal conductance over time. This study relied on ambient ozone concentrations, which may not reflect actual ozone flux into leaves. Finally, Wittig et al. (2007) assessed chronic effects across growing seasons, which can smooth out short-term fluctuations. In contrast, this study primarily captures acute responses under specific environmental conditions.

The quantity of the data causes some limitations. Only 4.88% of all available hourly values (722 hours) met the strict filtering criteria. This limited dataset made it difficult to achieve sufficient datapoints, especially at higher ozone levels. High ozone concentrations rarely occur within the 20–22°C range, meaning values above 120 µg/m³ could not be analysed. Solar radiation was also a limiting factor, as low ozone concentrations often coincided with low light levels, requiring model fitting for Amax estimation below 50 µg/m³.

Another limitation is the spatial mismatch between measurement locations. Ozone data were taken from Wekerom-Riemterdijk, approximately six kilometres from Loobos. This introduces uncertainty due to spatial variability.

This study also does not account for cumulative ozone exposure, focusing only on hourly instantaneous relationships. Therefore, chronic or delayed effects are not captured. This likely explains why no clear threshold was observed.

Another important factor is the interaction between radiation and stomatal behaviour. At high light intensity, stomata are more open, increasing ozone uptake and enhancing short-term sensitivity.

Although these limitations are important, the study still shows a clear negative relationship between ozone and GPP. The steep decline in Amax suggests strong short-term physiological responses. However, if averaged over longer timescales, the magnitude would likely be smaller.

Overall, this study suggests that tropospheric ozone reduces carbon uptake in Loobos, contributing to a positive feedback on atmospheric CO.

Conclusion

In this research, the effect of ozone on the GPP of Loobos was estimated. Hourly measurements of ozone were collected from monitoring station Wekerom-Riemterdijk, along with values for GPP, Rg, T, and VPD from the ICOS data portal for Loobos. These were half-hourly measurements which were converted to hourly values to temporally align them with the ozone values.

With this data, light response curves (LRCs) were constructed for different ozone concentration ranges, selecting only measurements where the temperature was in the range 20–22°C and VPD was between 0–15 kPa, to isolate the effect of ozone.

The results reveal a clear and consistent negative correlation. GPP, as measured by Amax in light response curves, declined by nearly 50% between ozone levels of 20–30 μg/m³ and 110–120 μg/m³.

In contrast to long-term studies such as Wittig et al. (2007), which found an 11% reduction in photosynthesis using seasonal averages and cumulative ozone uptake, this study captures short-term ozone impacts that are not smoothed over time.

The most plausible mechanism for the observed reduction is stomatal closure induced by ozone-triggered signalling, which directly limits CO uptake and photosynthetic efficiency. No threshold ozone concentration was identified, supporting the idea that stomatal responses are immediate and continuous rather than threshold-based.

However, the results also show that ozone impacts are strongly modulated by environmental conditions such as radiation and stomatal behaviour, suggesting that future studies should incorporate canopy conductance explicitly.

In conclusion, this study provides evidence that acute ozone exposure can strongly reduce GPP in Loobos. Trees show an average decline of nearly 50% between low and high ozone conditions. This indicates that ozone contributes indirectly to climate change by reducing the terrestrial carbon sink.